

Ruel Sinha

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SKILLS

Languages: Java, Python, C, JavaScript, HTML/CSS
Frameworks: React JSX, Node.js, Tailwind CSS
Dev Tools: Git, Google Cloud Platform, VS Code, Eclipse

EDUCATION

University of Waterloo Sep 2026 – Apr 2031 (*ongoing*)
Bachelor of Software Engineering (Honours, co-op) Waterloo, ON
• Admission average: **99.67%**

PROJECTS (viewable on ruelsinha.com)

ruelsinha.com | *React, Tailwind CSS, AWS S3* 2025 – Present
• Programmed and deployed website containing information and projects
• Used Tailwind CSS and Motion.dev to effectively style pages
• Used react.jsx to implement interactables and embedded files

Ball Trajectory analysis | *Differential equations* 2025
• Tracked and modeled the trajectory of a spinning ping pong ball as a function
• Used multivariable calculus and 2nd order non-homogeneous differential equations
• Matched trajectory while accounting for air resistance and spin

Remote Controlled Sumo Robot 2024
• Mechanically engineered and built a shell for a gearbox
• Used the torque of the gearbox's wheels to lift ramps on the robot's shell

3D LED Cube 2024
• Soldered and and pressed circuit board
• Programmed unique light pattern using Arduino IDE

Multiplayer Snake Game 2023
• Programmed unique features using python
• Engineered optimized code and runtime

EXPERIENCE

UWAT VEXU Robotics Design Team – Software Subteam Sep 2026 – Ongoing
University of Waterloo Waterloo, ON
• Engineer robotics software for the international VEX U robotics competition

Math Tutor 2024 – 2026
Vancouver Technical Secondary School Vancouver, BC
• Selected by school to tutor junior students in Math and Physics

Elections Officer Oct 2024
Elections BC Vancouver, BC
• Guided voters and carried out ballot processing

OTHER

CEMC Math Contests: Top 100s globally 2021 – 2026
SAT: 1570 2025
Math award & Science Award (Highschool) 2026
Piano: RCM level 6 2026